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Scuola di Ingegneria Industriale e dell'Informazione
Corso di Laurea Magistrale in Mathematical Engineering – Quantitative Finance

Computational Finance: Asset Allocation Project

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1 Constrained and Robust Efficient Frontier

1.1 Setup and Constraints

We construct the mean–variance efficient frontier using in-sample data from 2018 to 2022. The optimization is performed with the `Financial Toolbox` in MATLAB under the following constraints: full investment ($\sum_{i=1}^{N} w_i = 1$), no short selling ($w_i \geq 0$), a maximum weight per asset of 25% ($w_i \leq 0.25$), and exposure limits on macro groups (Defensive assets $\leq 40\%$, Cyclical + Defensive assets between 45% and 75%). Using these constraints, we generate an efficient frontier with 100 feasible portfolios and identify:

- **Portfolio A**: Minimum-Variance Portfolio (MVP)
- **Portfolio B**: Maximum Sharpe Ratio Portfolio (MSRP)

1.2 Resampled Efficient Frontier (Robust Portfolios)

To account for estimation error in expected returns and the covariance matrix, we re-estimate the efficient frontier using a resampling approach. The goal is to obtain a *robust* version of the frontier under the same constraints used in the nominal case.

1.2.1 Methodology

Starting from the in-sample estimates of the mean vector $\widehat{\mu}$ and covariance matrix $\widehat{\Sigma}$ computed from historical log-returns, the robust frontier is constructed via a resampling procedure inspired by Michaud's approach. The steps are:

1. For each resampling iteration $i = 1, \ldots, N_{\text{res}}$ (with $N_{\text{res}} = 100$), we generate a synthetic sample of T observations from a multivariate normal distribution: $R_t^{(i)} \sim \mathcal{N}(\widehat{\mu}, \widehat{\Sigma}), \quad t = 1, \ldots, T$.
2. From the simulated data, we compute new moment estimates: $\mu^{(i)} = \frac{1}{T} \sum_{t=1}^{T} R_t^{(i)}$, $\Sigma^{(i)} = \mathrm{Cov}\big(R^{(i)}\big)$.
3. The updated moments $(\mu^{(i)}, \Sigma^{(i)})$ are loaded into the `Portfolio` object, preserving all constraints (allocation bounds, sector limits, full investment). We then compute the constrained efficient frontier with $n_{\text{port}} = 100$ portfolios: $W^{(i)} = \{w_1^{(i)}, \ldots, w_{n_{\text{port}}}^{(i)}\}$.
4. After completing all resampling iterations, we average the weights across simulated frontiers: $\bar{w}_j = \frac{1}{N_{\text{res}}} \sum_{i=1}^{N_{\text{res}}} w_j^{(i)}$, $j = 1, \ldots, n_{\text{port}}$. The curve $\{\bar{w}_j\}_{j=1}^{n_{\text{port}}}$ defines the *robust efficient frontier*.

For each averaged portfolio $\bar{w}_j$, we finally compute annualized return, annualized volatility, and Sharpe ratio using the original in-sample estimates $(\widehat{\mu}, \widehat{\Sigma})$. In particular, the Sharpe ratio is defined in the same way as in the nominal case (annualized return divided by annualized volatility), but the inputs are evaluated using $(\widehat{\mu}, \widehat{\Sigma})$ rather than the simulated moments.
From this robust frontier we extract:

- **Portfolio C**: Robust Minimum-Variance Portfolio, is selected as the portfolio on the robust frontier with the lowest volatility: $j_C = \arg\min_j \sigma(\bar{w}_j), \qquad w_C = \bar{w}_{j_C}$.
- **Portfolio D**: Robust Maximum Sharpe Ratio Portfolio, is selected as the portfolio with the highest Sharpe ratio: $j_D = \arg\max_j \mathrm{Sharpe}(\bar{w}_j), \qquad w_D = \bar{w}_{j_D}$.

1.3 Full Comparison of portfolios A-D

Table 1 reports the weights of all four portfolios (nominal and robust). Table 2 summarizes their risk–return metrics.

Table 1: Weights for all portfolios: A (MVP), B (MSRP), C (Robust MVP), D (Robust MSRP).

Asset	**A (MVP)**	**B (MSRP)**	**C (Robust MVP)**	**D (Robust MSRP)**
1	0	0.2499	0	0.1074
2	0.0958	0.0001	0.0906	0.1063
3	0	0	0	0.0107
4	0.2500	0.2500	0.2500	0.1850
5	0.2367	0.2500	0.2231	0.1889
6	0	0	0	0.0235
7	0.1308	0	0.1289	0.0074
8	0	0	0	0.0085
9	0	0	0	0.0286
10	0	0	0	0.0326
11	0	0	0.0040	0.0334
12	0	0.2500	0.0032	0.1749
13	0.2192	0	0.2138	0.0455
14	0	0	0	0.0052
15	0.0675	0	0.0863	0.0325
16	0	0	0	0.0095

Table 2: Performance comparison of Portfolios A-D.

	A (MVP)	**B (MSRP)**	**C (Robust MVP)**	**D (Robust MSRP)**
Annualized Return	0.0568	0.0971	0.0565	0.0782
Annualized Volatility	0.1826	0.2049	0.1827	0.1959
Sharpe Ratio	0.3110	0.4738	0.3091	0.3992

Discussion

The comparison between the nominal and resampled portfolios highlights the impact of estimation uncertainty on portfolio optimization. Minimum-variance portfolios are inherently stable: the robust MVP (C) remains almost identical to the classical MVP (A), both in terms of weights and in-sample performance (Sharpe ≈ 0.31). This reflects the fact that the MVP depends primarily on the covariance matrix, which is relatively insensitive to sampling noise. In contrast, the nominal maximum-Sharpe portfolio (B) is highly concentrated, with several weights at the 25% upper bound. This behaviour illustrates the instability of Sharpe-maximising allocations with respect to small changes in expected returns. The resampled MSRP (D) produces a much smoother and more diversified allocation, but at the cost of lower efficiency: its expected return decreases (7.8% vs. 9.7%) and its Sharpe ratio is weaker (0.40 vs. 0.47). This trade-off is consistent with robust optimisation, which reduces sensitivity to estimation error but typically yields lower in-sample performance. Overall, resampling has only a marginal effect on the MVP but significantly improves the stability of the MSRP by preventing extreme weight concentration, resulting in allocations that are more balanced and more practical to implement.

2 Black-Litterman Model

The Black–Litterman model combines market-implied equilibrium returns with investor views to generate posterior expected returns. Using in-sample returns and their covariance matrix, we derive equilibrium returns from market-cap weights, introduce three linear views on selected assets/sectors, and compute the BL posterior. These updated returns are used to construct the efficient frontier (long-only, fully invested), from which we extract the Minimum Variance portfolio (E) and the Maximum Sharpe portfolio (F). The impact of the views on portfolio weights and efficiency is then evaluated.

2.1 Equilibrium Returns

We start from the market-cap portfolio provided in `capitalization_weights.csv`. Using in-sample simple returns (2018–2022), we compute annualized first and second moments and estimate the market's risk-aversion parameter: $\lambda = \frac{\mathbb{E}[R_m]-r_f}{\mathrm{Var}(R_m)}$. We obtained $\lambda \approx 2.26$. But we chose to keep $\lambda = 1.2$, as in the lectures. The implied equilibrium returns then follow from $\mu = \lambda \Sigma w_{\mathrm{mkt}}$, where Σ denotes the annualized covariance matrix of simple returns and w_{mkt} the market-capitalization weights. This defines the prior distribution: $\mu \sim \mathcal{N}(\mu,\ \tau\Sigma)$, $r \sim \mathcal{N}(\mu, \Sigma)$, with τ a scalar scaling the uncertainty of the prior mean.

2.2 Investor Views

Subjective views are expressed in the linear form $q = P\mu + \varepsilon,\ \varepsilon \sim \mathcal{N}(0, \Omega)$, where P contains the view portfolios, q the expected excess performance for each view, and Ω the associated uncertainty (taken diagonal, proportional to the variance of each view portfolio).
The following annualized views are incorporated into the BL model:

- **View 1:** cyclical assets outperform defensive assets by 2% annually.
- **View 2:** asset 3 outperforms asset 11 by 1%.
- **View 3:** asset 7 outperforms the average defensive asset by 0.5%.

Their uncertainty is modeled with the diagonal confidence matrix $\Omega_i = \tau\, P_i \Sigma P_i^\top,\ \tau = \frac{1}{T}$,

2.3 Posterior Returns and Efficient Frontier

The BL posterior expected returns and their uncertainty are

$$\mu_{\mathrm{BL}} = \left[(\tau\Sigma)^{-1} + P^\top \Omega^{-1} P\right]^{-1} \left[(\tau\Sigma)^{-1}\mu + P^\top \Omega^{-1} q\right],$$

$$\Sigma_{\mathrm{BL}} = \left[(\tau\Sigma)^{-1} + P^\top \Omega^{-1} P\right]^{-1}.$$

Consistently with the model, the predictive covariance used in portfolio optimization is $\Sigma_{\mathrm{opt}} = \Sigma + \Sigma_{\mathrm{BL}}$. Using (μ^{BL}, Σ_{BL}), we compute the efficient frontier under standard investment constraints, Table 3 reports the shift in expected returns implied by the Black–Litterman update, comparing the prior equilibrium returns with the posterior values incorporating the three investor views.

2.4 Portfolios E (Minimum Variance Portfolio) and F (Maximum Sharpe Portfolio

From the BL frontier we extract two portfolios:

- **Portfolio E (Minimum Variance)**: the lowest-risk point on the BL frontier.
- **Portfolio F (Maximum Sharpe)**: the portfolio maximizing $(\mu^{BL} - r_f)/\sigma$.

Table 3: Black–Litterman: Prior/Posterior Returns and Portfolio Weights

(a) Pre vs Post Expected Returns

Asset	Prior	BL
1	0.0638	0.0669
2	0.0407	0.0382
3	0.0553	0.0578
4	0.0362	0.0369
5	0.0429	0.0431
6	0.0614	0.0691
7	0.0505	0.0536
8	0.0568	0.0640
9	0.0679	0.0861
10	0.0569	0.0644
11	0.0497	0.0495
12	0.0534	0.0555
13	0.0518	0.0569
14	0.0539	0.0573
15	0.0433	0.0435
16	0.0572	0.0595

(b) BL Portfolio Weights (E and F)

Asset	MinVar	MaxSharpe
1	0	0.1627
2	0	0
3	0	0
4	0.7340	0.1263
5	0.2469	0
6	0	0.0242
7	0.0191	0.0750
8	0	0.1930
9	0	0.2185
10	0	0.2002
11	0	0
12	0	0
13	0	0
14	0	0
15	0	0
16	0	0

Effects of the Investor's Views on Weights and Efficiency

Impact on Expected Returns. The contribution decomposition shows that **View 1** is the main driver of the posterior adjustments. Its contributions are mostly positive and vary significantly across assets, with particularly large effects around assets 6, 7, 8, 9, 10 and 12. A few assets display small negative contributions. This reflects the fact that the view propagates through the covariance structure, rather than only affecting specific assets. **View 2** and **View 3** generate very small positive or negative adjustments spread across some assets. View 3 in particular causes small negative adjustments for several Defensive assets.
As shown in the left panel of Figure 1, the Black–Litterman posterior returns incorporate the three views in a smooth and correlation-consistent way, rather than producing sharp or localized changes in individual assets.

Impact on Maximum Sharpe Portfolio and on Minimum Variance Portfolio. Figure 1 highlights that the allocation changes once the views are incorporated. The BL Max Sharpe portfolio (right panel) increases its weight in Cyclical assets (especially 8, 9, 10) and reduces exposure to Defensive assets (such as 5), as expected from View 1. Asset 7 also receives a small positive adjustment due to View 3. The Minimum Variance allocation (center) changes

too, but with a lighter effect. Asset 7 has a negative adjustment, while Assets 4 (cyclical) and 5 (defensive) increase. It is important to note that the Minimum Variance portfolio reacts to the views only indirectly, since it depends exclusively on the covariance matrix and not on expected returns. BL views influence the solution only through the interaction between the updated covariance structure and the no-short-selling constraints. As a result, some assets that are not favored by the views (e.g., Asset 5) may still increase their weight if they contribute to lowering portfolio volatility, while others (such as Asset 7) may be reduced despite receiving a positive return adjustment.

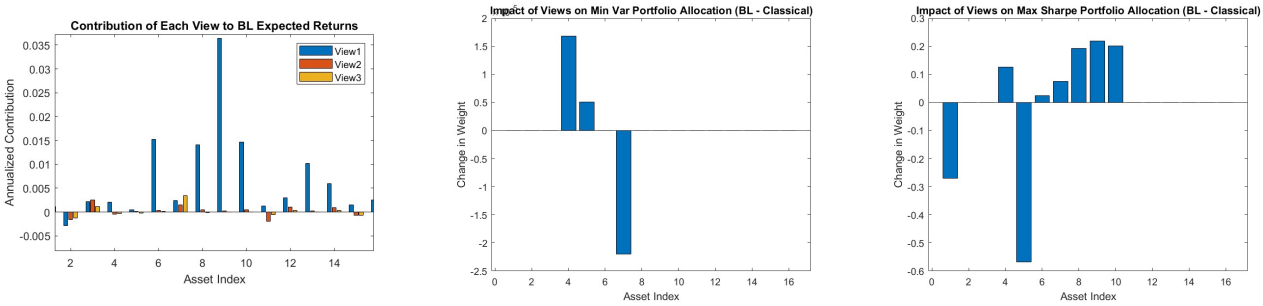


Figure 1: Impact of investor views on (left) expected returns, (center) Minimum Variance portfolio, and (right) Maximum Sharpe portfolio.

3 Diversification–Based Optimization

3.1 Portfolio G: Maximum Diversification Ratio

The first portfolio is constructed by maximizing the *Diversification Ratio* (DR), defined as

$$\mathrm{DR}(w) = \frac{w^\top \sigma}{\sqrt{w^\top \Sigma w}}, \tag{1}$$

where w is the vector of portfolio weights, σ is the vector of individual asset volatilities, and Σ is the covariance matrix of returns. The numerator represents the weighted average of asset volatilities, while the denominator is the total portfolio volatility. Maximizing the ratio therefore favours portfolios that combine assets with low correlation and heterogeneous risk profiles.

The optimization is carried out with MATLAB's `fmincon` by minimizing the negative diversification ratio, with portfolio weights constrained to lie between 0 and 0.20, summing to one, and with each macro-group required to represent at least 15% of the total portfolio weight.

The solution produces the weight vector $w^{(G)}$, from which we also report the achieved maximum diversification ratio. The results can be found in Table 4.

3.2 Portfolio H: Equal Risk Contribution (Risk Parity)

The second portfolio targets *risk parity*, i.e. an allocation in which each asset contributes equally to total portfolio risk. Given a weight vector w, portfolio volatility is

$$\sigma_p = \sqrt{w^\top \Sigma w}, \tag{2}$$

and the marginal risk contribution (MRC) of asset i is

$$\mathrm{MRC}_i = \frac{(\Sigma w)_i}{\sigma_p}, \tag{3}$$

leading to the absolute and relative risk contributions:

$$\mathrm{RC}_i = w_i \cdot \mathrm{MRC}_i, \qquad \mathrm{RC}_i^{(\mathrm{rel})} = \frac{\mathrm{RC}_i}{\sigma_p}. \tag{4}$$

Risk parity requires all relative contributions to be equal.
To enforce this, we minimize the dispersion of risk contributions using a *minimax* formulation. In practice, the objective supplied to MATLAB's `fminimax` computes the deviation of each asset's risk contribution from the target (their average), and the algorithm minimizes the ℓ_∞ norm of this deviation. This approach is more effective than `fmincon`, as the risk–parity objective is non-convex and highly sensitive to the initial guess; attempts with `fmincon` produced visibly uneven risk contributions, whereas `fminimax` achieves nearly perfect equality.
The optimization is carried out under the same constraints as in Portfolio G. As expected, all 16 assets have the same relative risk contribution, equal to 6.25%. The results can be found in Table 4.

3.3 Benchmark and Comparison

For comparison we also include the equally weighted portfolio ($w_i = 1/N$). All three portfolios (G, H, and EW) are evaluated in terms of Diversification Ratio, annualized volatility, Sharpe ratio and the effective number of assets $N_{\mathrm{eff}} = 1/\sum_i w_i^2$, which is the inverse of the Herfindahl index.
Portfolio G shows a concentrated allocation, in line with typical behaviour of Max–DR portfolios, only a limited subset of assets receives non-zero weights, which is consistent with the results obtained in our implementation.
Both the Risk Parity and Equally Weighted portfolios instead assign more homogeneous weights. This is explained by the fact that individual asset volatilities in the dataset are relatively similar, implying that equalizing risk contributions requires weights close to the uniform allocation. This is confirmed by the high effective number of assets ($N_{\mathrm{eff}} \approx 15.6$).
Finally, the Risk Parity portfolio achieves the highest Sharpe ratio among the three. This follows directly from its lower volatility, since risk–parity strategies effectively exploit diversification by distributing total risk across all assets.
The summary comparison statistics and the portfolio weights are reported in Table 5 and Table 4, respectively.

Table 4: Portfolio weights for Max-DR (G), Risk Parity (H), and Equally Weighted (EW) portfolios.

Asset	$\mathbf{w_G}$	$\mathbf{w_H}$	$\mathbf{w_{EW}}$
1	0	0.0506	0.0625
2	0.2000	0.0759	0.0625
3	0.1353	0.0580	0.0625
4	0.1958	0.0870	0.0625

Asset	$\mathbf{w_G}$	$\mathbf{w_H}$	$\mathbf{w_{EW}}$
5	0.0443	0.0738	0.0625
6	0	0.0530	0.0625
7	0.1895	0.0627	0.0625
8	0	0.0576	0.0625
9	0.2000	0.0505	0.0625
10	0	0.0578	0.0625
11	0.0351	0.0623	0.0625
12	0	0.0599	0.0625
13	0	0.0625	0.0625
14	0	0.0600	0.0625
15	0	0.0725	0.0625
16	0	0.0560	0.0625

Table 5: Comparison of Max-DR, Risk Parity, and Equally Weighted portfolios.

Portfolio	**Div. Ratio**	**Vol. (Ann.)**	**Sharpe**	$\mathbf{N_{eff}}$
Max Diversification Ratio (G)	1.251	0.2019	0.208	5.69
Risk Parity (H)	1.154	0.2050	0.292	15.62
Equally Weighted (EW)	1.153	0.2096	0.287	16.00

4 PCA and Conditional Value-at-Risk

In this section we use Principal Component Analysis (PCA) on the in-sample returns (Jan. 2018–Dec. 2022) to identify the main sources of common risk across the assets. By keeping only the components that explain most of the variance, we obtain a cleaner and less noisy representation of the covariance structure. Based on this PCA covariance, we construct then two portfolios: a maximum Sharpe ratio portfolio and a portfolio that minimizes CVaR_5% while respecting a volatility cap.

4.1 PCA and Number of Components Explaining 85% of Variance

PCA is a statistical technique that identifies the orthogonal directions (principal components) along which standardized returns vary the most. Given standardized data x_i, the projection onto a direction u is $x' = x^\top u$, and the variance preserved by this projection is $(x^\top u)^2$. Maximizing this variance over all unit vectors u yields the eigenvalue problem $Cu = \lambda u, \quad C = \frac{1}{n} X'^\top X'$, where C is the correlation matrix of standardized returns, λ is the variance explained by each principal component, and the associated eigenvectors define the loading directions. Sorting the eigenvalues in decreasing order therefore ranks components by importance.
The cumulative variance explained by the first k components is

$$\text{Explained}(k) = \frac{\sum_{i=1}^{k} \lambda_i}{\sum_{i=1}^{N} \lambda_i}.$$

We select the smallest k such that $\text{Explained}(k) \geq 0.85$. In our sample, the 85% threshold is reached at $k = 3$. Figure 2 displays the explained variance for each component.

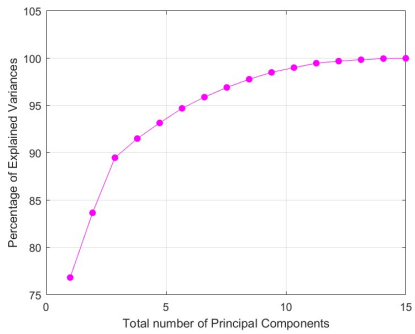


Figure 2: Percentage of variance explained by each principal component.

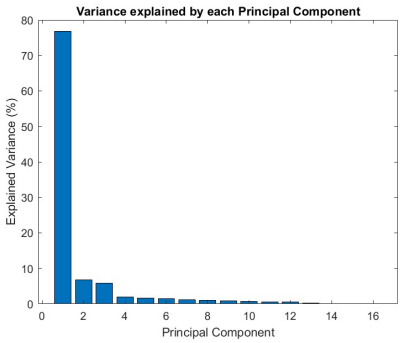


Figure 3: Variance by each Component

Let B be the loading matrix of the first k components and F the corresponding factor returns, with factor covariance $\Sigma_f = \text{cov}(F)$. Since PCA is performed on standardized data, we rescale back to original units using the diagonal matrix of volatilities $\Lambda = \text{diag}(\sigma_1, \ldots, \sigma_N)$ and mean vector μ. The reconstructed returns are $\widehat{R} = \mu + \Lambda(FB^\top)$, and the idiosyncratic component is $\varepsilon = R - \widehat{R}$, with diagonal variance matrix: $D = \text{diag}(\text{var}(\varepsilon))$. The resulting PCA-based covariance estimator is

$$\Sigma_{\text{PCA}} = \Lambda \left(B\Sigma_f B^\top + D \right) \Lambda.$$

4.2 Portfolio I – Maximum Sharpe Ratio with PCA Covariance

Using Σ_{PCA}, we compute the maximum Sharpe ratio portfolio under long-only constraints $0 \leq w_i \leq 0.25$ and full investment. In addition, exposure to the first eigenvector of the *sample* covariance matrix is capped: $w^\top v_1 \leq 0.5$. This controls concentration in the dominant source of systematic risk. The resulting portfolio achieves: Annual Sharpe Ratio = 0.4809.

4.3 Portfolio J – Minimum $\text{CVaR}_{5\%}$ with Volatility Cap

We minimize the historical $\text{CVaR}_{5\%}$ of portfolio returns:

$$\text{CVaR}_{0.05}(w) = -\,\mathbb{E}\big[r_t^\top w \mid r_t^\top w \leq \text{VaR}_{0.05}(w)\big],$$

subject to full investment, the bounds $0 \leq w_i \leq 0.25$, and the volatility cap

$$\sqrt{w^\top \Sigma_{\text{ann}} w} \leq 0.15.$$

In our setting, a 15% volatility target is not achievable. The minimum-variance portfolio satisfying the same weight constraints already has an annual volatility of 17.99%, so the cap is infeasible. As a result, the optimizer keeps the constraint active but converges to the closest feasible point on the boundary, yielding a solution with volatility approximately 0.18. The resulting portfolio achieves a final result of $\text{CVaR}_{5\%} = 0.027649$

Table 6: Portfolio Weights (I vs J)

Asset	$w^{(I)}$	$w^{(J)}$
1	0.25	0
2	0	0.1395
3	0	0
4	0.25	0.25
5	0.25	0.25
6	0	0
7	0	0.1191
8	0	0
9	0	0
10	0	0
11	0	0
12	0.25	0
13	0	0
14	0	0
15	0	0.2414
16	0	0

Table 7: Risk Tail-Risk Measures

Portfolio	$\text{CVaR}_{5\%}$	Vol.	MaxDD
Portfolio I	0.031875	0.20494	-0.28851
Portfolio J	0.027649	0.17993	-0.30488

4.4 Comparison between Portfolios I and J

Portfolio I, constructed using the PCA-based covariance with neutrality to the first principal component, achieves an annual volatility of 20.45% and a daily $\text{CVaR}_{5\%}$ of 0.0319. This behavior is coherent with a maximum Sharpe portfolio as it is I: the allocation takes more risk in order to maximize the return-to-risk ratio, which results in higher volatility and larger tail losses.

Portfolio J, instead, is obtained by directly minimizing $\text{CVaR}_{5\%}$ under the same weight constraints and a volatility cap. As expected, its CVaR of 0.0265 is lower, and its volatility of 17.30% is also significantly smaller. The trade-off is visible in the maximum drawdown: Portfolio J shows a drawdown slightly larger (-29.65%) than Portfolio I (-28.75%), reflecting its more conservative and less factor exposed structure.
Overall, portfolio I delivers higher risk-adjusted performance but at the cost of greater volatility and tail losses. Portfolio J achieves a clear reduction in CVaR and risk at the expense of lower expected performance and a slightly deeper worst-case drawdown.

5 Personal Allocation Strategy

We design a personal allocation strategy based on a statistical factor model estimated via Principal Component Analysis (PCA). The objective is to construct a long-only portfolio under realistic allocation constraints, maximizing a mean-variance objective evaluated on a "cleaned" covariance matrix and assessing its performance over the out-of-sample period (January 2023 - November 2024).

5.1 Methodology

Let $\widehat{\Sigma}$ denote the annualized in-sample covariance matrix of log-returns (2018–2022) and $\widehat{\mu}$ the corresponding vector of annualized expected returns. We compute the eigen-decomposition

$\widehat{\Sigma} = V\Lambda V^\top$, where V is the matrix of orthonormal eigenvectors and $\Lambda = \mathrm{diag}(\lambda_1, \ldots, \lambda_N)$ contains the eigenvalues sorted in descending order. We retain the first $k = 3$ principal components and define

$$F = V_{(:,1:k)}, \qquad \Sigma_f = \mathrm{diag}(\lambda_1, \lambda_2, \lambda_3),$$

so that F collects the loadings on the main systematic factors and Σ_f is the factors covariance matrix.
For each asset i, the variance explained by the first k PCs is $\mathrm{Var}_i^{\mathrm{expl}} = \sum_{j=1}^{k} \lambda_j F_{ij}^2$, and the idiosyncratic variance is the residual $\sigma_{\varepsilon_i}^2 = \widehat{\Sigma}_{ii} - \mathrm{Var}_i^{\mathrm{expl}}$. Collecting these terms on the diagonal of a matrix D yields the idiosyncratic component. The factor-model covariance is then $\Sigma_{\mathrm{FM}} = F\Sigma_f F^\top + D$, which provides a denoised and more stable estimate of the risk structure.

Given $\widehat{\mu}$ and Σ_{FM}, we consider the mean–variance objective

$$f(w) = \widehat{\mu}^\top w - \gamma\, w^\top \Sigma_{\mathrm{FM}} w,$$

with risk-aversion parameter $\gamma = 1$. The weights w are obtained by solving

$$\max_w \; f(w) \quad \text{s.t.} \quad \sum_{i=1}^{N} w_i = 1, \qquad 0 \le w_i \le 0.20 \; \forall i,$$

so that the portfolio is fully invested, long-only, and each position is capped at 20%. The problem is solved numerically in `MATLAB` using `fmincon` on the negative of $f(w)$, with the equally-weighted portfolio as starting point.
The optimal weight vector is then applied to the out-of-sample price series (January 2023 – November 2024), with a semi-annual rebalancing back to the target weights. From the resulting portfolio value process we compute annualized return, annualized volatility, Sharpe ratio, maximum drawdown and Calmar ratio.

5.2 Optimal Weights

Table 8 reports the optimal weights returned by the optimization. The solution is relatively concentrated, with positive allocations on six out of sixteen assets and binding upper bounds on four of them.

Asset	1	2	4	5	12	16
Weight	0.20	0.139	0.20	0.20	0.20	0.061

Table 8: Optimal portfolio weights of the personal allocation strategy (remaining assets have 0%).

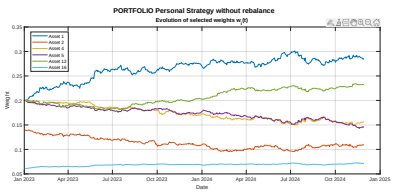
5.3 Out-of-Sample Performance

On the validation period from 1 January 2023 to 30 November 2024, starting from an initial portfolio value of $V_0 = 100$, we implement a semi-annual rebalancing scheme that restores the weights to their target levels. Rebalancing every six months is a common industry practice, as it controls drift in portfolio exposures while limiting transaction frequency and turnover costs. Table 9 reports the main performance indicators computed over the out-of-sample period.

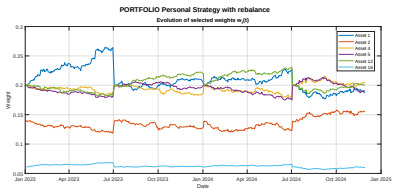
	Ann Return	Ann Volatility	Sharpe	Max DD	Calmar
Value	0.2024	0.1097	1.8440	-0.0923	2.1918

Table 9: Out-of-sample performance metrics of the personal allocation strategy.

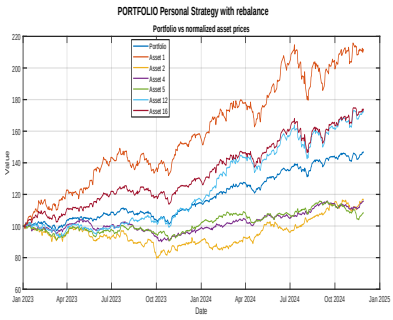
Figure 4 summarizes the impact of rebalancing and the resulting portfolio performance. The left and middle panels compare the evolution of portfolio weights under two scenarios: without rebalancing, where weights drift according to relative asset performance (left), and with semi-annual rebalancing, which restores the target allocation (middle). The right panel shows the price evolution of all individual assets in the universe together with the cumulative value of the optimized portfolio. The comparison highlights the smoothing and diversification effects achieved by the allocation strategy. At the same time, the strong performance observed in this period should be interpreted with caution, as the evaluation window is relatively short (approximately two years) and largely characterized by a broadly monotonic upward trend across most assets. A more comprehensive assessment would require testing the portfolio under market conditions featuring higher dispersion or downturns, such as episodes of stress or recession.



(Left) Without rebalancing.



(Middle) With semi-annual rebalancing.



(Right) Assets vs. optimized portfolio.

Figure 4: Portfolio weight dynamics with and without rebalancing (left and middle panels) and evolution of individual assets versus the optimized portfolio (right panel).

5.4 Discussion

The proposed allocation is intentionally simple: it relies on a PCA-based factor model to separate systematic and idiosyncratic risk, then applies a standard mean-variance criterion under realistic allocation constraints. The use of only the first few principal components stabilizes the covariance estimate by filtering out high-frequency noise in the correlation structure, while the idiosyncratic matrix D preserves asset-specific risk.

The resulting portfolio is quite concentrated, reflecting the fact that mean-variance optimization, even with a cleaned covariance matrix, tends to allocate more capital to assets with the most favourable risk–return trade-off under the model. Nevertheless, out-of-sample results show limited volatility and drawdowns relative to the achieved return, suggesting that the PCA-based risk model has been effective in controlling estimation error and avoiding extreme, unstable allocations.

6 Final Discussion

Table 10: In-sample performance of portfolios A–J. Returns and volatilities are annualized.

Ptf	Return	Volatility	Sharpe	Max DD	Calmar
A	0.0568	0.1826	0.3110	-0.3069	0.1851
B	0.0971	0.2049	0.4738	-0.2885	0.3365
C	0.0565	0.1827	0.3091	-0.3081	0.1832
D	0.0782	0.1959	0.3992	-0.3216	0.2432
E	0.0666	0.1731	0.3846	-0.2560	0.2601
F	0.0582	0.2333	0.2493	-0.4060	0.1432
G	0.0419	0.2019	0.2077	-0.3656	0.1147
H	0.0599	0.2050	0.2923	-0.3504	0.1710
I	0.0971	0.2049	0.4738	-0.2885	0.3366
J	0.0609	0.1799	0.3384	-0.3049	0.1997

Table 11: Out-of-sample performance of portfolios A–J. Returns and volatilities are annualized.

Ptf	Return	Volatility	Sharpe	Max DD	Calmar
A	0.1429	0.1000	1.4294	-0.1023	1.3978
B	0.2208	0.1238	1.7838	-0.0925	2.3873
C	0.1431	0.0998	1.4333	-0.1021	1.4014
D	0.1745	0.1073	1.6253	-0.0943	1.8509
E	0.0767	0.0976	0.7857	-0.1134	0.6762
F	0.1949	0.1237	1.5751	-0.0957	2.0368
G	0.1781	0.1141	1.5602	-0.0951	1.8722
H	0.1922	0.1132	1.6977	-0.0999	1.9240
I	0.2209	0.1238	1.7839	-0.0925	2.3871
J	0.1172	0.0976	1.2014	-0.0997	1.1762

6.1 Out-of-Sample Performance of Portfolios A–J

Table 11 reports the out-of-sample performance of portfolios A–J over the validation period (January 2023 – November 2024). Portfolios **B** (classical Max-Sharpe) and **I** (PCA-based Max-Sharpe) deliver the strongest out-of-sample results, with the highest Sharpe ratios and relatively mild drawdowns. The PCA exposure constraint has little impact in this regime, as the two portfolios exhibit almost identical allocations and performance. Minimum-variance portfolios **A** and **C** continue to behave nearly identically, confirming the intrinsic stability of minimum-variance weights. The robust Sharpe portfolio **D** achieves slightly lower volatility than B but also materially lower returns, resulting in a weaker Sharpe ratio. Black-Litterman portfolios show mixed behaviour: the conservative version **E** performs poorly, while the return-oriented version **F** is stronger but still does not match the efficiency of B and I. Diversification-based portfolios **G** (Max-DR) and **H** (Risk Parity) provide balanced and stable performance, with moderate returns, controlled volatility, and shallow drawdowns. Though less aggressive than Max-Sharpe solutions, they offer a solid risk–return trade-off. The CVaR-driven portfolio **J** attains low volatility but sacrifices too much return, leading to a substantially lower Sharpe ratio. Its improved tail-risk metrics do not translate into superior drawdown outcomes during

this calm period.
Finally, it is important to note the difference between in-sample and out-of-sample risk-adjusted performance. In-sample (2018-2022) Sharpe ratios are relatively low (0.2-0.5), whereas out-of-sample (2023-2024) Sharpe ratios increase markedly (0.8-1.9) and volatilities decline. The validation window is comparatively calm and upward-trending, so realised Sharpe ratios are higher than those implied by the in-sample moments. Overall, the out-of-sample evidence suggests that, because the validation period was calm and upward-trending, aggressive portfolios (B and I) performed extremely well, while defensive or tail-risk-focused portfolios showed weaker results simply because the market environment did not reward their caution.

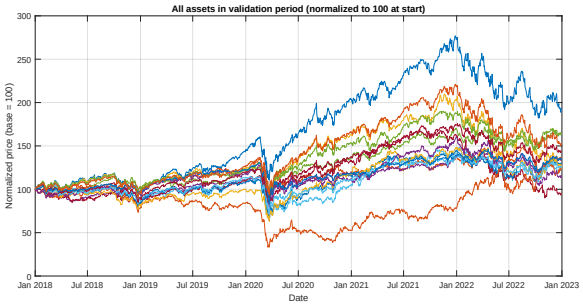


Figure 5: In-sample period (2018–2022).

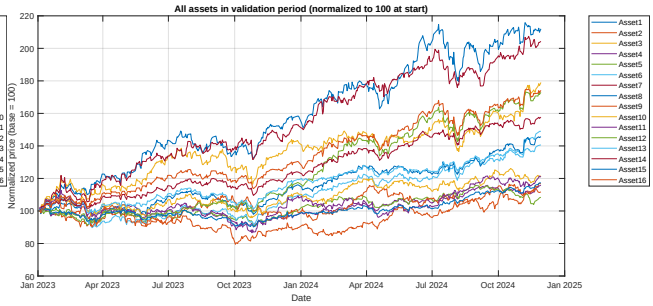


Figure 6: Out-of-sample period (2023–2024).

Figure 7: Normalised price evolution of all assets in the in-sample and out-of-sample periods.

6.2 Risk-Adjusted Performance and Robustness

A portfolio is considered robust if its behaviour remains consistent across the in-sample (2018–2022) and out-of-sample (2023–2024) periods, and if performance evolves coherently with market conditions rather than reflecting noise or overfitting.

Minimum-variance and diversification-based portfolios (A, C, G, H). Portfolios A and C show almost identical volatility, Sharpe ratios, and drawdowns in both samples, reflecting the natural stability of minimum-variance allocations. Portfolios G and H also behave consistently across regimes: although their in-sample Sharpe ratios are modest, they improve smoothly out of sample in line with the calmer market environment and maintain balanced, diversified profiles, indicating strong robustness.

Maximum-Sharpe portfolios (B, D, I). Portfolios B and I achieve the highest out-of-sample Sharpe ratios, confirming the strong performance of moderately cyclical, return-seeking allocations in this period. Their similarity shows that the PCA constraint in I has limited effect in this regime. Portfolio D, while more diversified and less concentrated, loses too much expected return relative to B and I, illustrating the typical trade-off of robustification: improved stability at the cost of reduced in-sample efficiency. Aggressive Max-Sharpe allocations (B, I) benefit the most from the favourable market environment.

Black–Litterman portfolios (E, F). Portfolio E performs reasonably in-sample but becomes one of the weakest portfolios out of sample, suggesting that its posterior returns did not align with realised market conditions. Portfolio F performs better but remains below the Max-Sharpe solutions in terms of risk-adjusted performance.

Tail-risk portfolio (J). Portfolio J achieves lower volatility and CVaR but gives up too much return. In the absence of significant tail events during the validation period, these benefits do not translate into better overall performance.

6.3 Impact of Cyclical / Neutral / Defensive Exposures

Cyclical Exposures Portfolios B and I (both with 50% Cyclical exposure) achieve the strongest performance, confirming that a moderate cyclical tilt combined with some Defensive allocation enhances risk-adjusted results. Portfolio E (73% Cyclical) retains low volatility and limited drawdowns in-sample, but its efficiency collapses out of sample. Portfolio F (90% Cyclical, minimal Defensive) is structurally fragile: in-sample it shows the highest volatility, the deepest drawdown and one of the weakest Sharpe ratios; out-of-sample its performance improves in the favourable market environment, but its Sharpe ratio remains below the best cyclical portfolios, confirming that excessive cyclical concentration raises risk without delivering superior efficiency.

Defensive Exposures Portfolios with high Defensive exposure (especially J with 63%) show lower volatility but also lower expected returns. Portfolios A, C, G, and H (35–40% Defensive) maintain stable but not outstanding profiles, with meaningful drawdowns despite their defensive tilt.

Neutral Exposures Neutral weights act primarily as diversification buffers. Portfolios with sizeable Neutral components (A, C, H) display intermediate behaviour, neither strongly pro-cyclical nor strongly defensive.

Out-of-Sample Perspective The validation period reinforces the patterns observed in-sample. Moderately cyclical portfolios B and I remain the strongest performers. Extremely cyclical portfolio F continues to show high returns but also elevated volatility and weaker Sharpe ratios. Portfolios with meaningful Defensive exposure (A, C, J) preserve their characteristic stability but lag behind in efficiency, particularly J. Portfolio E represents a clear regime mismatch: despite its strong cyclical tilt, it becomes one of the weakest strategies out of sample.

Table 12: Macro-group exposures for portfolios A–J.

Portfolio	Cyclical	Neutral	Defensive
A	0.25	0.35	0.40
B	0.50	0.25	0.25
C	0.25	0.35	0.40
D	0.36	0.28	0.35
E	0.73	0.02	0.25
F	0.90	0.10	0.00
G	0.40	0.22	0.38
H	0.30	0.30	0.40
I	0.50	0.25	0.25
J	0.25	0.12	0.63